

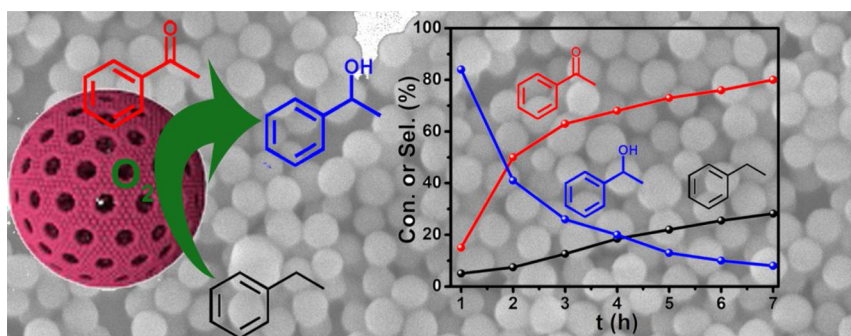
Study of doped mesoporous ceria with tailored oxygen vacancies for enhances activity for ethylbenzene oxidation application

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GRAPHICAL ABSTRACT



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ABSTRACT

Herein, we present a ligand-assist induction self-assembly method to prepare mesoporous CeO_2 (mCeO_2) nanospheres. Then, a series of mCeO_2 catalysts with Ni or Co ions doped with various Ce/Ni (Co) ratios were prepared by this method. These mCeO_2 :xNi/Co (mCeO_2 with different molar Ni or Co ions doping) samples were studied using a range of characterization techniques (XRD, N_2 adsorption-desorption, SEM, TEM, XPS, H_2 -TPR, O_2 -TPD and Raman spectroscopy). Both mCeO_2 and mCeO_2 :xNi/Co nanospheres show high surface areas. XPS spectra reveal that the ratios of $\text{Ce}^{3+}/\text{Ce}^{4+}$ increase as the doping concentration of Ni/Co ions increasing, which indicated that mCeO_2 :xNi/Co samples are more reducible than pure mCeO_2 . The H_2 -TPR, O_2 -TPD, and Raman analysis results reveal that the doping of Ni or Co ions significantly increased the concentration of surface oxygen vacancies in mCeO_2 . As for ethylbenzene oxidation reaction, doping-mesoporous ceria catalysts exhibited a superior catalytic activity when compared to pure mesoporous ceria catalysts prepared by the same method.

1. Introduction

As we all know, cerium (Ce) is one of the most abundant elements (66.5 p.p.m.) among the Earth's crust, and is much more abundant than that of tin (2.3 p.p.m) or copper (60 p.p.m) [1–4]. With a high abundance, CeO_2 has been wide applications in the chemical field such as

luminescence [5], fuel cells [6], organic catalysis [7] and photocatalytic [8]. Ce has two common valence states cerium (III) and cerium (IV) due to its electronic configuration $[\text{Xe}] 4f^2 6s^2$, which give ceria remarkable chemical and physical properties [3]. This unique properties result in CeO_2 product numerous oxygen vacancy defects during many reactions. It has been confirmed, the generation and migration of oxygen vacancy

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